

ECONOMETRICS I

Problem Set 4

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Case study: Cigarettes and baby birth weight

- Figure 1 shows the relationship between a baby's birth weight (in ounces) and the number of cigarettes smoked by the mother (in logarithm).
 - What observation would, if removed, have the largest impact on the estimated slope? Indicate in the graph and justify.
 - What observation would, if removed, have the largest impact on the estimated constant? Indicate in the graph and justify.
 - How would the observations be distributed if the logarithm were not applied to the x-axis? Draw a graph representing the resulting relationship.

Figure 1: Birth weight vs. maternal cigarette consumption

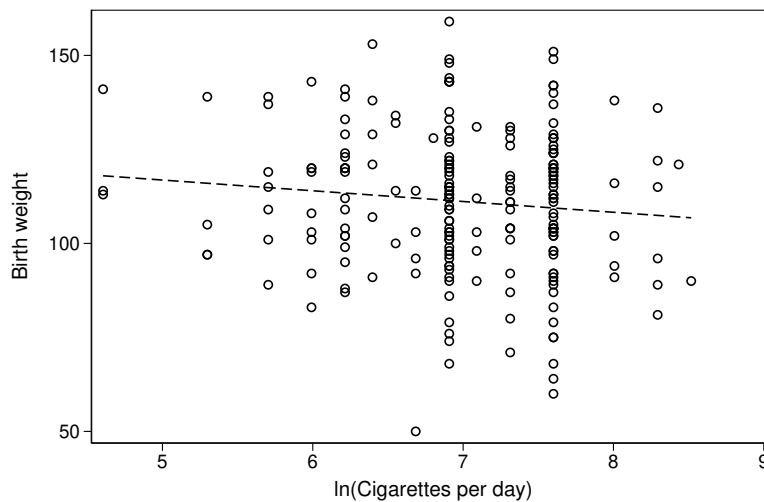


Table 1: OLS Regressions

Variable dep.: Birth weight (in ounces)						
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	117.0*** (1.049)	115.4*** (3.107)	115.1*** (2.906)	118.1*** (3.500)	138.1*** (18.76)	138.5*** (16.27)
Cigarettes per day	-0.463*** (0.0916)	-0.486*** (0.0926)	-0.582*** (0.109)	-0.589*** (0.111)		
Household income	0.0928** (0.0292)			0.0538 (0.0367)	0.169? (0.106)	0.202* (0.0922)
Mother's education years		0.331 (0.233)		-0.438 (0.320)	-1.172 (0.889)	-1.102 (0.786)
Father's education years			0.413 (0.213)	0.494 (0.283)	1.156 (0.752)	
ln(Cigarettes per day)					-4.426* (2.098)	-2.680 (1.833)
Observations	1388	1387	1192	1191	161	212
R^2	0.030	0.024	0.030	0.033	0.070	0.036

Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 1 shows a series of estimates of the effect of maternal tobacco consumption on newborn weight. Answer the following related questions.

- How does the computer calculate the coefficients in table 1? Indicate the meaning of each symbol you use.
- How does the computer calculate the standard errors in table 1? Indicate the meaning of each symbol you use.
- What is the expected weight of a baby when the mother does not smoke and the household has an income level equal to 30?
- What does the note with $p < 0.05$ refer to? What does p mean? Explain extensively and be very precise.
- Any star in the household income coefficient in (5) has been replaced by a question mark. What number (if any) of stars should be on it?
- Which of the estimated constants has a higher level of significance?
- In table 2 (see last page) a value appears with the symbol '???'. What is the corresponding value?
- Indicate which of the following violations of assumptions would have an invalidating effect on the calculation of table 2:
 - Regressors correlated with the error term.

- (b) Autocorrelation of errors.
- (c) Non-normal distribution of errors.
10. Approximately, what is the 95% confidence interval of the impact of daily cigarettes according to (4)? Consider all the information given in the table.
 11. How does the impact of tobacco consumption differ between regressions (4) and (5)?
 12. Why does household income appear as a significant variable in (1) but not significant in (4)?
 13. (5) has a better R^2 than (4). Nevertheless, the coefficient associated with smoking is more significant in (4). How can this apparently contradictory difference be understood?
 14. In (4), both household income and years of education of mother and father appear non-significant. Does that mean they don't matter? How could you see if they matter jointly? State the null hypothesis in matrix terms and indicate what the test statistic would be.

Table 2: Variance-covariance matrix of $\hat{\beta}$ from regression 1 of table 1

Constant	Cigarettes per day	Household income	
1.1004	-0.030930	-0.025694	Constant
	???	0.00046254	Cigarettes per day
		0.00085193	Household income